

| <b>B.Sc. (Entire Computer Science)-I, Level - 4.5 UG Certificate Level</b> |                                                                                     |    |            |                |               |    |
|----------------------------------------------------------------------------|-------------------------------------------------------------------------------------|----|------------|----------------|---------------|----|
| <b>Sem:</b>                                                                | II                                                                                  |    |            |                |               |    |
| <b>Paper Category:</b>                                                     | DSC3-2 (Major) (Group-I)                                                            |    |            |                |               |    |
| <b>Paper Name:</b>                                                         | Introduction to Microcontroller and Embedded System<br><b>Paper Code : G06-0203</b> |    |            |                |               |    |
| <b>Credit:</b>                                                             | 02                                                                                  |    |            | <b>Theory:</b> | 2 Hrs./Week   |    |
| <b>Marks:</b>                                                              | <b>UA:</b>                                                                          | 30 | <b>CA:</b> | 20             | <b>Total:</b> | 50 |

### Course Objective:

1. To develop specialists in hardware-software co-design for application specific Electronic systems.
2. To prepare students for demonstrating the acquired knowledge.
3. To encourage the student to develop skills for accepting challenges of upcoming technological advancements.

**Course Outcomes:** Upon successful completion of this course, students will be able to

1. Design, test and critically evaluate embedded solutions to real-world situations using digital components (sequential and combinational).
2. Recognize the key features of embedded systems in terms of computer hardware and be able to discuss their functions. You will be aware of the key factors affecting computing hardware evolution.
3. Design, test and critically evaluate embedded solutions to real-world situations using (embedded) computer systems interfaced with digital hardware

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|-------------------------------------------------|---------------------------|-------------------------------|
| <b>Unit I: Introduction to Microcontroller:</b> | <b>No. of Lectures:15</b> | <b>Weightage: 12-18 Marks</b> |
|-------------------------------------------------|---------------------------|-------------------------------|

**Fundamentals of Microcontroller:** Introduction to 8051 Microcontroller, Difference between Microprocessor and Microcontroller, Overview and features of MCS-51 Family, Salient features and block diagram of 8051, Pin description, Internal memory, Registers and SFRs, Port structure, Clock and Reset circuit. Timer/counter, Serial port, Instruction set of 8051:-

Instruction set, Instruction types - Data transfer, Arithmetic, Logical instruction execution and timing. Simple programs based on ports using Embedded C (LED on/off, Port in Port Out)

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|--------------------------------------------------|---------------------------|-------------------------------|
| <b>Unit II: Introduction to Embedded System:</b> | <b>No. of Lectures:15</b> | <b>Weightage: 12-18 Marks</b> |
|--------------------------------------------------|---------------------------|-------------------------------|

Definition of an embedded system, Types of Embedded System, Basic architecture of embedded system, Concept of Hardware and Software Design in Embedded System, characteristics of Embedded System, Advantages and Application of Embedded System., Introduction to Embedded c and Development Tools: Structure of Embedded C, I/O port programming, Development tools for Embedded System: Introduction to Keil Micro vision, Steps involved in Programming with Keil Micro Vision simulation. Introduction to flash magic, Steps involved in programming of the microcontroller. Basic Concepts of Designing of microcontroller based embedded system

#### **Reference Books:**

1. Muhammad Ali Mazidi, Janice Gillispie Mazidi, Rolin D. McKinley's, "The 8051 microcontroller and Embedded systems using Assembly and C", Second Edition Pearson
2. Kenneth Ayala, Delmar Cengage Learning, "The 8051 Microcontroller Architecture, Programming and Applications" , Third Edition TMH
3. Raj Kamal, Embedded systems Architecture, Programming and Design TMH